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Understanding virus filtration membrane performance

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ABSTRACT

Virus filtration membranes are used to obtain virus clearance during the purification of biopharmaceutical products. These direct flow (also referred to as dead end or normal flow) filtration membranes are designed to reject virus particles and yield >98% product recovery for proteins less than 170 kDa. Virus filtration feed streams generally have high purity and high product concentrations. Decrease in permeate flux during virus filtration, which reduces filter capacity, is most likely due to fouling by small aggregates of the product species present in the feed stream.

Retrovirus and parvovirus filtration membranes, as well as two ultrafiltration membranes with NMWCOs of 300 and 10 kDa (kg mol^{-1}), were tested using feed streams consisting of minute virus of mice (MVM) in the presence and absence of 1% (w/v) BSA. A novelty of our work comes from the fact that we directly compare the performance of these membranes using realistic model feed streams consisting of MVM (an FDA recommended model virus) and BSA. This approach provides an industrially relevant benchmark of the engineering performance of virus filters with respect to a limited set of structural and operational variants. All asymmetric membranes were operated in direct flow, constant pressure mode with the filtration surface (i.e. skin surface) facing away from the feed stream as is industrial practice for virus filtration. In addition, the 10 kDa ultrafiltration membrane was also operated with the skin surface facing the feed stream as is industrial practice for ultrafiltration. While little rejection of virus particles was observed for the 10 kDa membrane when the skin surface faced away from the feed stream, significant (in excess of 3 log) virus rejection was observed when the skin surface faced the feed stream.

Changes in permeate flux with filtrate volume were determined along with parvovirus rejection. Decreases in permeate flux were shown to result from fouling in the presence of BSA. No decrease in permeate flux was observed in the absence of BSA. The decrease in permeate flux has been analyzed in terms of classical pore blockage models. The results of this work, which compare the performance of virus filtration and similar ultrafiltration membranes, provide insights into designing virus filtration membranes.

1. Introduction

Obtaining virus clearance data is essential in the manufacture of protein-based biopharmaceuticals, as well as in the plasma-derived blood products industries [1–5]. Many therapeutic proteins from mammalian cell culture fermentations are in large-scale production worldwide. Manufacturers are required to validate virus clearance by methods such as size exclusion removal or inactivation before regulatory authorities will approve a product for release to market.

During the purification process, manufacturers of monoclonal antibodies typically demonstrate reduction of 10^3 to 10^5 or more virus particles than is estimated in a single dose equivalent of the unprocessed bulk. Depending on the antibody titer, estimates of the number of virus particles in a single dose equivalent could be as high as 10^{10} to 10^{15} retrovirus-like particles per mL[1]. Since viruses cannot be grown to such a high titer, even if a single unit operation could achieve 10^{15} fold virus reduction, it could not be validated.

Virus inactivation technologies include physical methods (e.g., heat and radiation) and chemical methods (e.g., solvent detergents, low pH). Virus removal technologies, on the other hand, include filtration (by size exclusion) and ion-exchange chromatography. The level of virus clearance by inactivation or removal in a purification train is calculated by summing the clearance obtained from individual unit operations. Virus reduction factors from two unit operations with the same mechanism of action may not be added

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together. In addition, due to the high level of variability in the infectivity assays used to quantify the level of virus clearance, reduction factors of less than 10-fold may not be included in the overall reduction factor.

This work focuses on virus removal by filtration. Virus filtration membranes are designed to reject virus particles while allowing the product of interest to pass through the membrane pores. These membranes are large pore ultrafiltration membranes with a very narrow pore-size distribution. Virus removal by filtration is easily scalable and robust as it depends on size exclusion [6]. Further, validation of parvovirus clearance by inactivation is difficult given the resistance of these virus particles to inactivation methods such as low pH hold steps [7]. Virus filters originally were operated in tangential flow mode [8]. Tangential flow filtration is more appropriate when the membrane is to be cleaned and reused a number of times and when the total particulate load is high. Virus filtration feed streams are relatively pure and have a high product concentration. Furthermore, the quantity of particulate matter present (i.e. percentage by volume of virus particles in the feed) is low. In addition, reuse of virus filters is not practical given the need to develop a validated cleaning protocol. Finally, the simplicity of operation combined with lower capital costs has resulted in virus filtration being conducted today using disposable direct flow filters [9].

Optimal virus filters must maximize capacity, throughput, and selectivity [10]. The capacity of a virus filter is the total volume of filtrate that can be processed before the flux declines to an unacceptably low value during constant pressure filtration. Throughput refers to the speed at which the feed can be filtered (maximum sustainable permeate flux). Selectivity refers to the ability to yield high product recovery and high virus particle retention. These filters must be able to process the entire bulk feed at acceptable filtrate fluxes, reject virus particles, and maximize protein passage [11]. The presence of even a few larger pores in a virus filter could lead to passage of virus particles. Thus, while virus filters are essentially large pore (300 kDa or larger NMWCO) ultrafiltration membranes, they have a very narrow pore-size distribution compared to ultrafiltration membranes. Virus filters typically consist of composite membrane structures made from hydrophilic polyethersulfone, polyvinylidene fluoride, and regenerated cellulose [12,13]. Unlike ultrafiltration membranes, asymmetric virus filtration membranes are operated with the filtration surface away from the feed inlet (i.e. skin side down).

Virus filtration typically is conducted with process streams that contain few impurities and high product protein concentrations. Virus filters contain pores that are only slightly larger than large proteins. For example, parvovirus retention filters must reject virus particles as small as 20 nm. Thus, the main foulant in virus filtration is the product protein and possibly its aggregates. Syedain et al. [11] have compared the capacity of asymmetric virus filtration membranes with the filtration surface towards and away from the feed stream. Their experiments were conducted using bovine serum albumin (BSA) solutions in the absence of virus particles. They indicate that the capacity is significantly higher when the filtration surface is away from the feed inlet, as is used in practice, due to the internal flow branching that can occur around blocked pore intersections.

Ultrafiltration membranes, on the other hand, are operated with the filtration surface (i.e. skin surface) facing the feed inlet. Ultrafiltration feed streams often contain much higher impurity loads than feed streams common to virus filtration membranes. These former membranes are cleaned-in-place and reused a number of times; therefore, it is essential to minimize irreversible fouling. Syedain et al. [11] indicate that when the membrane is used with the filtration surface facing the feed stream, flux decline is dominated by osmotic pressure effects and gel layer formation; whereas, when the membrane is used with the filtration surface away from the feed stream, flux decline is mainly due to irreversible fouling within the porous support structure of the membrane. In the case of asymmetric virus filtration membranes, virus particles often are trapped irreversibly within the more open support structure and the interconnections among the pores provide alternative flow paths for the fluid, thus minimizing the level of decrease in filtrate flux. This mode of operation protects the filtration surface from fouling, thus maximizing capacity and flux. (In the case of essentially symmetric virus filtration membranes, virus particle may be trapped within the pore structure of the membrane though there is no downstream skin surface to protect from fouling.) Interestingly, Guerra et al. [14] use membranes with the filtration surface away from the feed stream for tangential flow microfiltration in the dairy industry. Again the membrane support structure is used to protect the filtration surface from foulant deposition. As these membranes are reused a number of times, frequent back-pulsing of the permeate is required in order to minimize irreversible fouling of the support structure.

Though numerous studies have focused on the capacity, throughput, and selectivity of ultrafiltration membranes [15–18], few studies have focused on the performance of virus filtration membranes. Bohonak and Zydney [19] have shown that compaction of several commercially available virus filtration membranes during operation can lead to a decrease in permeate flux. However, by using selected asymmetric membranes with the filtration surface away from the feed stream, the support structure acts as a depth pre-filter minimizing flux decline. In another study, Bakhshayeshi and Zydney [20] investigated the effect of feed pH on protein transmission in the absence of virus particles. They found that membrane capacity and protein yield were a minimum at the protein isoelectric point, though the sieving coefficient was a maximum. Further, the yield and capacity were greater when asymmetric membranes were operated with the filtration surface away from the feed stream.

Unlike previous studies, we have investigated the performance of an asymmetric retrovirus and an essentially symmetric parvovirus filtration membrane and two asymmetric ultrafiltration membranes under conditions typically used in industry for virus filtration. The two ultrafiltration membranes have NMWCOs of 300 and 10 kDa. Feed streams consisted of minute virus of mice (MVM) particles in buffer. In addition, virus containing feed streams were spiked with BSA (1% (m/v) final concentration). Results were analyzed in terms of classical pore blockage models.

2. Materials and methods

2.1. Cell culture

A9 Mouse (*Mus musculus*) Fibroblast Cells (ATCC #CCL-1.4, Manassas VA) were used in these experiments. A frozen cryovial containing A9 cells from ATCC was thawed rapidly in a water bath at 37 °C. After thawing, the cells were added to a 15 mL conical tube containing 10 mL of fresh culture medium [High glucose Dulbecco's Minimal Essential Medium (DMEM) containing L-glutamine (Hyclone, Logan, UT) with 10% heat inactivated fetal bovine serum (FBS, Hyclone)]. The contents were centrifuged at 1000 rpm for 5 min, the supernatant was discarded, and 10 mL of fresh medium was added to the pellet. The resuspended cells were transferred to a 25 cm^2 cell culture filter flask and incubated at 37 °C and 10% CO_2. Cells were observed daily using an inverted microscope until they reached greater than 90% confluence, typically within 2–3 days.

Upon reaching the desired confluence, the cell culture medium was discarded from the vessel, and cells were rinsed with 5 mL Dulbecco's phosphate buffered saline (DPBS) solution, pH 7.4 (Invitrogen, Carlsbad, CA). The DPBS solution was discarded and 1.5 mL of 0.05% trypsin, 0.53 mM EDTA (Invitrogen) was added to the ves-

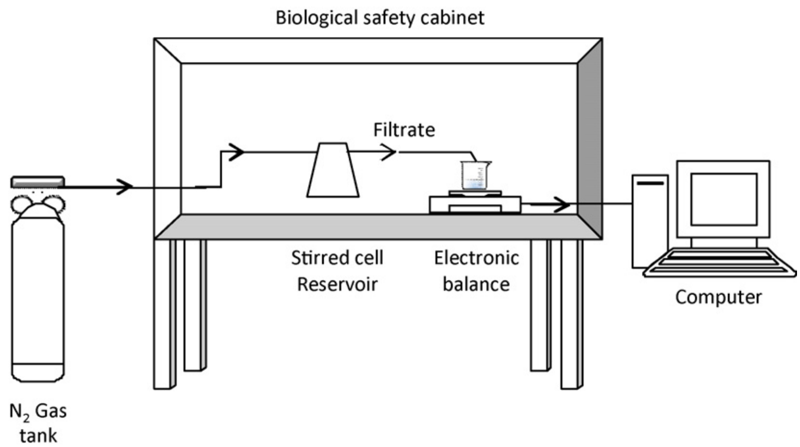

Fig. 1. Filtration experimental set up.

sel. The vessel was rocked at room temperature until cells began to round up as observed by the inverted microscope after approximately 60 s. Next, 1 mL of trypsin-EDTA was removed, leaving 0.5 mL in the vessel. The vessel was rocked every 30 s until the cells began to detach. Cells were rinsed from the bottom of the flask by aspirating 5 mL of fresh culture medium over the detaching monolayer. The suspension was aspirated several times to dispense the cells, and they were transferred to a 15 mL conical tube. A 100 μL sample of the suspension was removed in order to determine the cell density by staining with 0.4% Trypan blue dye (Hyclone). The cells were counted using a hemacytometer. About 5×10^6 cells were used to start a new culture in a 75 cm^3 cell culture filter flask or 1×10^7 cells for a 150 cm^3 cell culture filter flask. Cells were incubated, observed for confluence, and expanded into multiple vessels using the method for detachment described previously, except increasing reagent volume proportionally to the vessel size.

2.2. Virus propagation

One culture vessel was trypsinized as described previously to obtain a cell count using the Trypan blue staining method. The cell count was used to determine the amount of virus to dilute in 1 mL of culture medium to achieve a multiplicity of infection (MOI) of 0.001. Stock aliquots of minute virus of mice (MVM) (ATCC # VR-1346) were obtained from −80 °C storage, and thawed rapidly in a water bath at 37 °C. The media was discarded from all cell culture vessels, and each flask was inoculated with 1 mL of pre-diluted virus (per 75 cm^3 cell culture filter flask). The vessels were incubated at 37 °C and 10% CO_2 for 90 min, with gentle rocking approximately every 10 min. After 90 min adsorption, the vessel was rinsed with 3 mL of DPBS and 20 mL of fresh culture media was added. The vessels were incubated at 37 °C and 10% CO_2 and monitored daily with an inverted microscope for cytopathic effects (CPE) and pH. pH was buffered by adding small drops of 7.5% sodium bicarbonate solution (Invitrogen) to the medium until it remained orange to red in color. Virus infected medium was harvested at the height of CPE, typically 10 days after infection. The culture medium was frozen and thawed three times by freezing in the −80 °C freezer and rapidly thawing in a 37 °C water bath. Medium was placed in 50 mL centrifuge tubes and centrifuged at 5000 rpm for 5 min to remove cell debris. Virus stock was stored at −80 °C.

2.3. Virus filtration

Frozen MVM stock virus was thawed rapidly in a 37 °C water bath, and spiked into high glucose DMEM containing L-glutamine solution to a concentration of 10^8 virus particles/mL. These solutions are referred to as low protein solutions. In order to determine the effect of high protein loads in the feed stream, bovine serum albumin (BSA) (Sigma–Aldrich, St. Louis, MO) was suspended in phosphate buffered saline (PBS) pH 7.4 to a concentration of 1% (w/v). MVM stock virus was spiked into the solution to a concentration of 10^8 virus particles/mL. These solutions are referred to as high protein solutions. The high protein solution was sterile filtered through a 0.2 μm sterile filter unit (Nalgene, Rochester, NY) to remove large protein aggregates.

Filtration experiments were conducted using feed streams consisting of low protein and high protein solutions containing 10^8 virus particles/mL. The solution was filtered in a standard 200 mL, Model 8200 Amicon stirred cell (Millipore, Billerica, MA) in direct flow filtration mode at a constant pressure of 0.2 MPa in a biological safety cabinet. Fig. 1 is a schematic representation of the filtration set up. Table 1 summarizes the various membranes tested. All membranes were single sheets (unlike commercial products), placed with the filtration surface away from the feed stream for the asymmetric membranes and with a non-woven support (Freudenberg and Co., Weinheim, Germany) underneath to prevent damage

Table 1
Membranes tested. Membrane specification data were provided by Pall Corporation, Port Washington, NY [22], and Millipore, Billerica, MA [21].

Trade name	Ultipor VF grade DV20	Omega 300	Omega 10	RVRM
Manufacturer	Pall	Pall	Pall	Millipore
Membrane	Hydrophilic modified PVDF	Modified polyethersulfone	Modified polyethersulfone	Polyethersulfone
Comments	Direct flow virus filtration membrane commercially used as a double layer, ≥3 log reduction for virus (>20 nm), ≥6 log reduction for virus (>50 nm), operating pressure of 0.1–0.3 MPa	Tangential flow ultrafiltration membrane, 300 kDa nominal molecular weight cut off, maximum operating pressure of 0.4 MPa	Tangential flow ultrafiltration membrane, 10 kDa nominal molecular weight cut off, maximum operating pressure of 0.4 MPa	Direct flow virus filtration membrane commercially used as a triple layer, >6 log reduction for virus (>80 nm (Phi 6)), approximate maximum operating pressure of 0.35 MPa

to the membrane from the stirred cell. A similar procedure has been used by previous investigators [11,19,20].

Additional virus filtration experiments using the 10 kDa ultrafiltration (Omega 10) membrane were conducted using a Model 8050, Amicon stirred cell (nominal feed volume 50 mL). The operating conditions used were identical to those used for the Model 8200 Amicon stirred cell. Four experimental conditions were investigated. The membrane was used skin side facing towards and away from the feed stream. Feed streams containing MVM in the presence and absence of BSA were used.

For both stirred cells it was observed that leakage of the feed solution around the O ring used to seal the membrane in the stirred cell occurred for asymmetric membranes when the membranes were oriented such that the support structure faced the feed stream. Leakage was prevented by applying parafilm (Pechiney Plastic Packaging Company, Chicago, IL, USA) on the lower screw thread of the Amicon stirred cell. In order to verify that leakage around the O ring had been successfully arrested, permeate fluxes were determined for all the membranes tested here with both surfaces towards and away from the feed stream. The permeate fluxes were identical showing no dependence on membrane orientation indicating that flow around the O ring had been arrested when the filtration surface was placed away from the feed stream for the asymmetric membranes.

Approximately 500 μL aliquots, for virus and protein analysis, were taken from the filtrate stream for every 10 mL permeate collected. Filtrate fluxes were recorded using a top loading balance to weigh the filtrate.

2.4. Virus quantification

The virus concentration in the cell culture for all experimental results was determined by real time polymerase chain reaction (RT-PCR). DNase master mix [water, DNase buffer, and DNase (Bio-Rad, Hercules, CA)] (9 μL) was pipetted into each well in a 96-well plate, then 1 μL of sample was added, and the plate was covered. (*Note*: each sample was tested in duplicate.) The plate was spun in a centrifuge until the velocity reached approximately 800 rpm, then immediately stopped. The plate was incubated in a 37 °C water bath for 40 min. SYBR green reagent, forward primer, and reverse primer (Bio-Rad) were thawed rapidly by hand. The PCR master mix was prepared in a microcentrifuge tube by adding 10 μL SYBR green reagent, 8.2 μL water, 0.4 μL forward primer, and 0.4 μL reverse primer to the tube for every reaction. Nineteen microliters of PCR master mix was added to each well in a fresh 96 well plate, and 1 μl of DNase treated sample was added to the PCR master mix. Eight standards, ranging from 10^2 copies/μL to 10^9 copies/μL, were thawed and included with each plate. The plate was analyzed in a Bio-Rad iCycler iQ5 RT-PCR machine.

2.5. BSA assay

Protein concentration was measured using a BCA Protein Assay Kit (Pierce, Rockford, IL) following the manufacturer's instruction. Using a 96-well microplate (Nalge Nunc International, Rochester, NY), 25 μL of unknown sample or standard albumin were added to the wells. Next 200 μL of working reagent was added to each well. The plate was covered and incubated at 37 °C for 30 min. After cooling to room temperature, the absorbance of each sample at 562 nm was measured using a microplate spectrophotometer (Benchmark Plus Microplate Spectrophometer, Bio-Rad). As described by the manufacturer, the protein concentration is determined and reported with reference to a standard albumin solution provided by the manufacturer. All samples were analyzed in triplicate and average values reported. For feed streams spiked with 1% (w/v) BSA, the majority of the protein present was BSA.

2.6. Critical point drying for scanning electron microscopy

Scanning electron microscopy was used to image both surfaces and the membrane cross section. To prevent pore collapse, critical point drying was performed prior to analyzing samples with a JEOL field-emission scanning electron microscope (JSM-6500F, JEOL Ltd., Tokyo, Japan). Small membrane samples were placed into specimen holder baskets. Baskets were placed directly into the liquid transfer boat filled with absolute ethanol, and then into the critical point drying apparatus. Cold tap water was run to the jacket of the chamber, after which the chamber was filled with liquid carbon dioxide and allowed to sit for 5–10 min. With the vent valve slightly open to maintain the liquid carbon dioxide level, the drain valve was opened to remove the absolute ethanol for approximately 3–5 min. The flushing action was repeated a least 7 times, until the absolute ethanol had completely displaced the water.

After flushing was complete, the chamber was again filled with liquid carbon dioxide. The temperature of the chamber was increased to approximately 36–38 °C by replacing the cold water in the jacket with warm water. The carbon dioxide gas was slowly vented off. Samples were removed from the chamber and the baskets, and mounted on microscope stubs. Samples were then coated with gold. Membrane cross sections were obtained by placing the membrane sample in liquid nitrogen for 10 min. The membranes were then broken and the resulting cross section imaged.

3. Results and discussion

Fig. 2 presents FESEM images of the surfaces of the four membranes tested. The two virus filtration membranes have a much more symmetrical structure than the two ultrafiltration membranes. However, while the Ultipor VF grade DV20 (DV20) is considered a symmetric membrane by the manufacturer, the RVRM is an asymmetric membrane. Asymmetric virus filtration membranes are designed such that the support structure, which is the upstream membrane surface, has a slightly larger pore size than the filtration surface and acts as an in-line prefilter that removes any product protein aggregates that could foul the filtration surface [6,10]. In the case of the DV20, though the porosity of the upstream surface appears lower, the nominal pore size of the two surfaces is essentially the same. The RVRM membrane is designed to retain retroviruses (greater than 50 nm in size). It has a larger pore size than the DV20 membrane which is designed to reject parvovirus particles, around 20 nm in size.

Fig. 2 also contains FESEM images of the cross-section of the four membranes tested. Obtaining the entire cross-section in a single image is difficult given the different membrane thicknesses. Consequently, the cross-sectional images are taken at different magnifications. Breaking the membranes after immersion in liquid nitrogen will lead to damage to the exposed pore structure. Nevertheless Fig. 2(c), (i) and (l) clearly shows the asymmetric structure of the RVRM, Omega 300 and Omega 10 membranes. The dense skin layer is supported by a much more open structure. Fig. 2(f), the cross-section of the DV 20 membrane, is clearly different. There is no support structure and the morphology appears the same throughout the cross section. The membrane seems essentially symmetric.

The two ultrafiltration membranes appear to have filtration surface (skin surface) pore-size distributions that are much broader than that of the virus filtration membranes. The Omega 300 appears to have a similar nominal filtration surface pore size as the DV20, although the DV20 has a lower porosity. The Omega 10 has a very tight filtration surface that is difficult to visualize clearly by FESEM. Nevertheless, it appears that there are a few relatively large pores

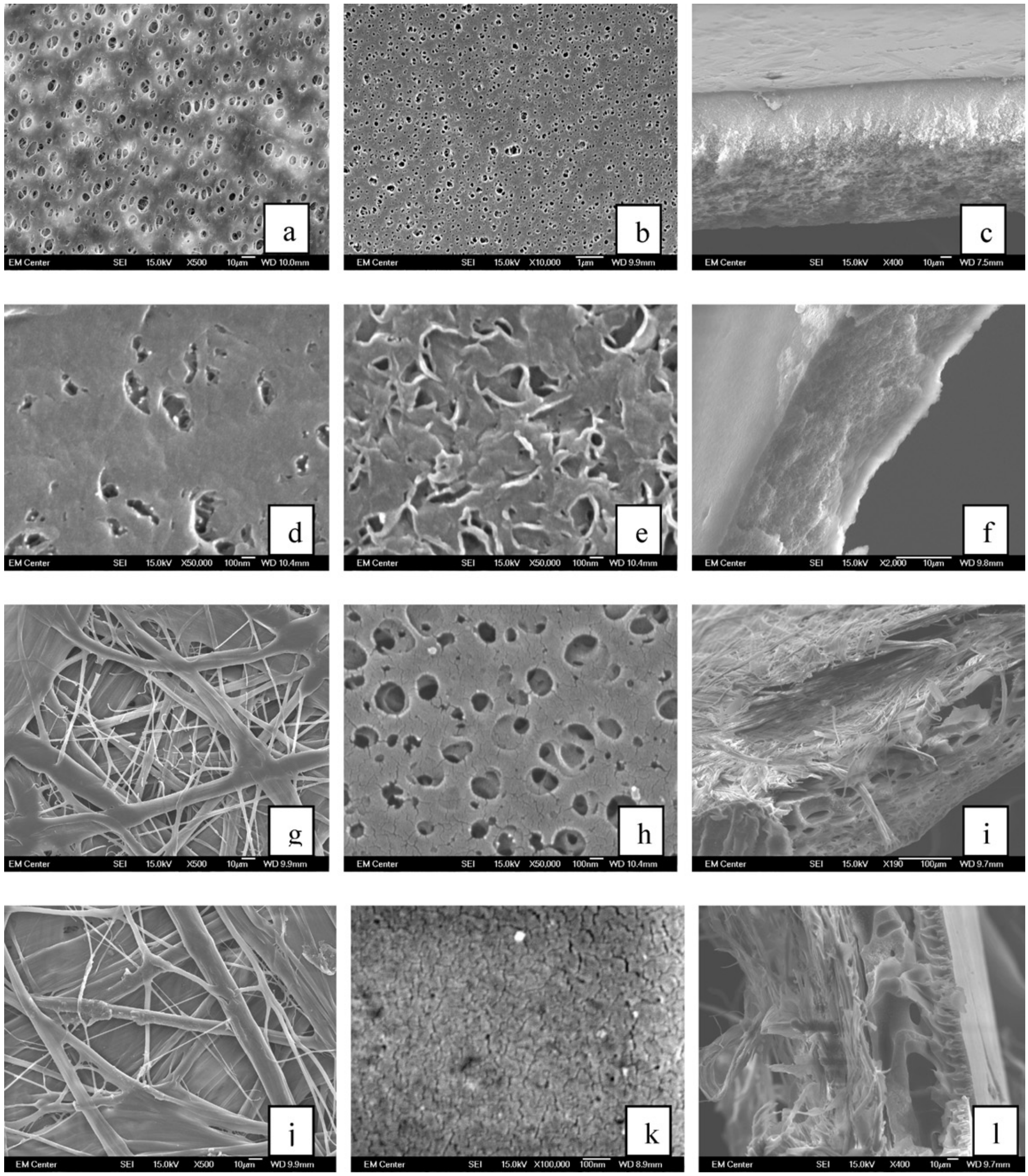

Fig. 2. SEM images of support side (upstream surface) of RVRM (a), DV 20 (d), Omega 300 (g) and Omega 10 (j) membranes at a magnification of 500 times, except for the DV 20 which is shown at a magnification of 50,000 times. SEM images of the skin surface (downstream surface) of the RVRM (b), DV 20 (e) Omega 300 (h) and Omega 10 (k) membranes are shown at a magnifications of 10,000 times RVRM, 50,000 times DV 20 and Omega 300 and 100,000 times Omega 10. Cross-sectional images are shown for RVRM (c) at 400 times, DV 20 (f) at 2,000 times, Omega 300 (i) at 190 times and Omega 10 (l) at 400 times magnification.

that could give rise to a broader pore-size distribution compared to the DV20.

Figs. 3–6 give results for membranes tested using the Amicon Model 8200 stirred cell with the membrane skin surface facing away from the feed for the asymmetric membranes. Fig. 3 gives water fluxes at 0.2 MPa. The RVRM and Omega 300 have the same water flux. According to the Millipore product information [21] the RVRM membrane has a permeability of $100\,\mathrm{L\,m^{-2}\,h^{-1}\,psi^{-1}}$, which gives a water flux of $3300\,\mathrm{L\,m^{-2}\,h^{-1}}$ at 0.2 MPa. Bohonak and Zydney [19] found the PBS permeability for the Omega 300 to be approximately 4×10^{-12} m at 0.1 MPa, where the permeability is defined as

$$L_p = \frac{\mu J_v}{\Delta P} \tag{1}$$

In Eq. (1), L_p is the membrane permeability, μ is the solution (in this case, PBS) viscosity, J_v the permeate flux and ΔP the pressure drop across the membrane. For the experimental conditions used here, the predicted water flux would therefore be $2880\,\mathrm{L\,m^{-2}\,h^{-1}}$. Both results are in excellent agreement with the water fluxes determined here. Fig. 2 indicates that even though the Omega 300 has a smaller nominal pore size than the RVRM membrane, the combination of porosity, pore-size distribution, and thickness results in a similar permeate flux as the RVRM membrane.

Pall product information [22] indicates that the water flux for the Omega 10 is $0.7–0.9\,\mathrm{mL\,min^{-1}\,cm^{-2}}$ at 0.37 MPa. This water flux results in a water flux estimate of about $300\,\mathrm{L\,m^{-2}\,h^{-1}}$ under the experimental conditions used here. Bohonak and Zydney [19] found the permeability of the DV20 to be about 0.025×10^{-12} m. Using Eq. (1), the predicted water flux under the experimental conditions

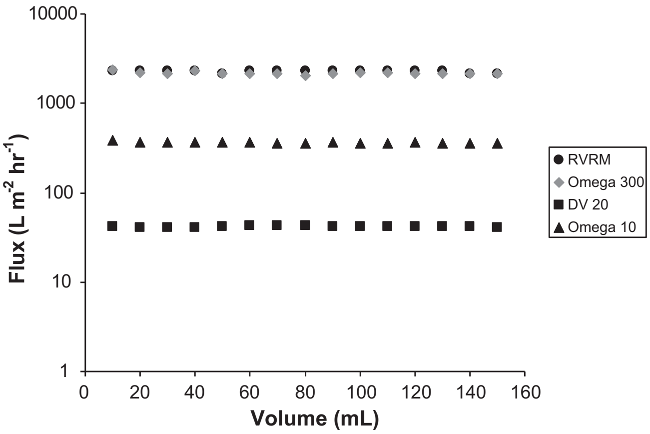

Fig. 3. Water fluxes at a pressure differential of 0.2 MPa.

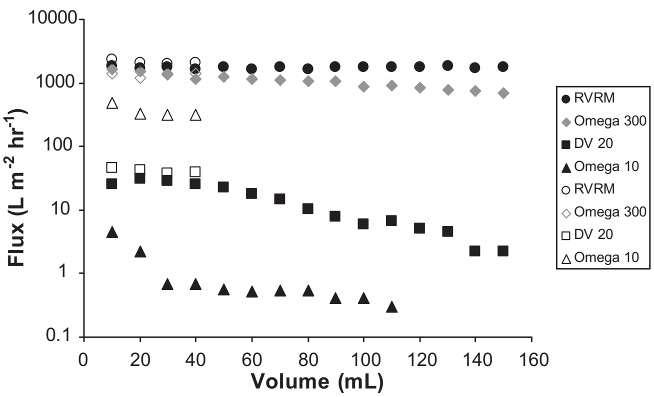

Fig. 4. Permeate fluxes for MVM feed streams containing 1% (w/v) BSA (filled symbols) and in the absence of spiked BSA (open symbols).

used here is 20 L m^2 h^{-1}. Both results are in excellent agreement with the water fluxes determined here. Although the Omega 10 appears to have a smaller nominal pore size than the DV20, it has a higher water flux at 0.2 MPa, due to the same combination of geometric characteristics mentioned previously.

Fig. 4 gives permeate fluxes for virus-loaded feed streams with (filled symbols) and without (open symbols) 1% BSA. In the presence of BSA, there is a decrease in permeate flux with volume processed for all four membranes. The RVRM, Omega 300 and DV20 were run until about 150 mL of permeate were collected. In the case of the Omega 10, only about 110 mL of permeate were collected before filtration was stopped due to the very low permeate flux. The initial

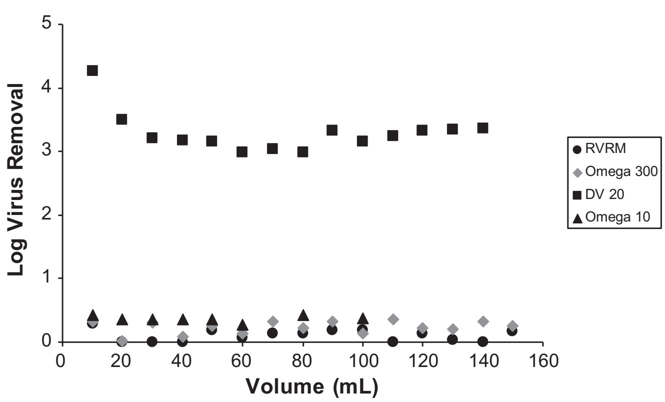

Fig. 5. Log virus removal for feed streams containing 1% (w/v) BSA.

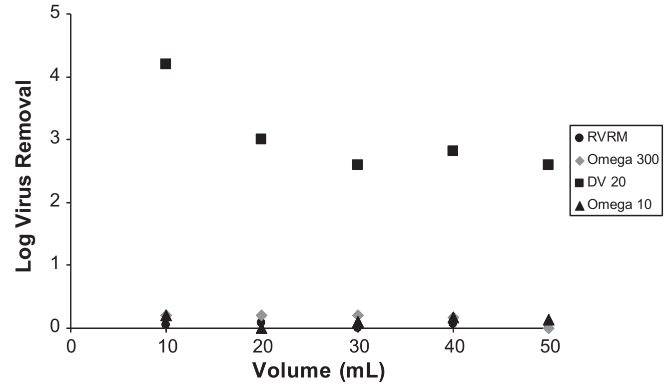

Fig. 6. Log virus removal for feed streams without addition of BSA.

feed volume was 200 mL in all cases. In addition, with the exception of the RVRM membrane, after removal of 40 mL of permeate the membranes displayed a decrease in permeate flux.

The open symbols in Fig. 4 give the variation of permeate flux in the absence of BSA. Here, removal of 40 mL of permeate from an initial feed volume of 200 mL does not lead to a decrease in permeate flux. Thus, the decrease in permeate flux is attributed to the presence of BSA and its concentration in the feed reservoir over time. Millipore product information [21] indicates that using the RVRM membrane will lead to more than 98% product recovery. Pall product information [22] indicates that the DV20 displays high transmission of proteins up to 160 kDa. Further Pall product information indicates that when used in tangential flow mode with the filtration surface upstream, the Omega 300 and 10 membranes will effectively retain proteins larger than 900 and 30 kDa respectively. Thus, little rejection of BSA is expected when using the Omega 300; whereas, almost complete rejection of BSA is expected for the Omega 10 during tangential flow filtration.

In this work, passage of 1% BSA, MW 70 kDa, was observed for all four membranes. Further, all BSA feed streams were prefiltered using a 0.2 μm sterile filter to remove large protein aggregates. Table 2 summarizes the observed BSA transmission after removal of 100 mL of permeate. As expected, no BSA rejection was observed for the RVRM, DV20 and Omega 300 membranes. However, significant rejection of BSA was observed for the Omega 10. Table 2 indicates that BSA rejection for the Omega 10 was around 0.25 when operated with the skin surface away from the feed stream. However, in additional experiments conducted using the small Amicon 8050 stirred cell with the membrane oriented such that the skin surface faced the feed stream (as is industrial practice for ultrafiltration), the BSA rejection was about 99% after removal of 30 mL of permeate.

The greater capacity of asymmetric virus filtration membranes operated with the filtration surface away from the feed stream has been attributed to removal of product protein aggregates by the membrane support structure acting as a depth filter [11,20]. In the case of the symmetric DV20 membrane, the pore size is essentially

Table 2
Average observed BSA transmission after removal of 100 mL permeate from a feed volume of 200 mL containing 1% (w/v) BSA where the membrane was oriented such that the skin surface was downstream. Transmission for the Omega 10 membrane after removal of 30 mL of permeate from a feed volume of 60 mL where the membrane was oriented such that the skin surface faces the feed stream is also included.

Membrane	Concentration in permeate/initial feed
RVRM	>0.98
Omega 300	>0.98
DV 20	>0.98
Omega 10	~0.75
Omega 10	~0.01 (skin surface facing the feed)

uniform. While protein aggregates can still be removed by the internal structure of the membrane, there is no downstream 'filtration surface' to protect. The results obtained here for the RVRM, DV20 and Omega 300 agree with this observation, as it is likely that protein aggregates that passed through the 0.22 μm sterile filter fouled those membranes. Further, the cross-sectional FESEM images indicate that for the three asymmetric membranes the skin surface has the smallest pore size. For the DV 20 it appears there is no significant change in the pore size through the membrane cross section.

The decrease in permeate flux for the Omega 10 is much more significant than the other three membranes. This membrane is designed to reject proteins with molar masses similar to BSA; thus, BSA rejection is expected. Since the Omega 10 shows significant rejection of BSA, it is likely that monomeric BSA, in addition to protein aggregates, foul the membrane. As indicted by Fig. 4, virus particles in the absence of BSA (i.e. low protein concentration) appear to cause little membrane flux decline (fouling). The initial virus concentration was 10^8 virus particles per mL or approximately 10^{-12} M; whereas, the BSA concentration was approximately 1.4×10^{-4} M or 8 orders of magnitude higher than the virus concentration. Thus, the much higher concentration of product protein indicates that small aggregates and perhaps dimers and trimers of the protein are the most likely component in the feed stream to cause membrane fouling. Furthermore, given that virus particles are much larger and more rigid than protein molecules, entrapment of a virus particle in the membrane support structure is unlikely to completely block a given pore (or intersection) allowing liquid to flow around the trapped particle.

Additional experiments were conducted with the Omega 10 membrane (skin surface away from the feed) using the Amicon 8050 stirred cell to verify this conclusion. We found that feed streams containing 1% (w/v) BSA in the presence and absence of virus particles displayed the same permeate flux as a function of permeate volume. Thus, there appears to be little interaction between the BSA and MVM particles. This result is not unexpected as MVM and BSA will be negatively charged at the feed pH (7.4)

Figs. 5 and 6 give log removal of virus versus permeate volume in the presence and absence of 1% BSA. The DV20 was the only explicit parvovirus retention filter tested. It shows about 3 log removal of MVM in the permeate as claimed by the manufacturer. Noteworthy is that all membranes were tested as a single layer, while the DV20 is sold commercially as a multilayer device that is likely to give a higher virus retention than the single layer tested here. Comparing Figs. 5 and 6, the LRV in the presence of BSA is a little higher. However given the PCR assay used here has an accuracy of about ±0.3 log units, the LRV in the presence and absence of BSA is the same within experimental error.

Figs. 5 and 6 indicate low levels of clearance of MVM by the RVRM, Omega 300 and Omega 10 membranes. Since the RVRM membrane is a retrovirus retention membrane that has a pore size larger than MVM, no MVM rejection was expected. Fig. 2(h) indicates that the Omega 300 also has filtration surface pores that are larger than MVM, leading to low log removal of virus. Comparing Figs. 5 and 6 there appears to be some rejection (much less than 1 log) for the RVRM and Omega 300 in the presence of BSA though this is within the uncertainty of the PCR assay.

Previous investigators have considered the use of ultrafiltration membranes for concentration of virus particles. Wickramasinghe et al. [23] investigated concentration of human influenza virus particles, which have an average size ~100 nm. They detected no virus particles in the permeate from a 100 kDa membrane but detected a very low virus concentration in the permeate from a 300 kDa membrane. Geraerts et al. [24] and Baekelandt et al. [25] used 100 kDa ultrafiltration membranes to concentrate lentiviral vectors, which are about 80–130 nm in size. Grzenia et al. [26,27] investigated concentration of densonucleosis virus using 300, 100, 50 and 30 kDa

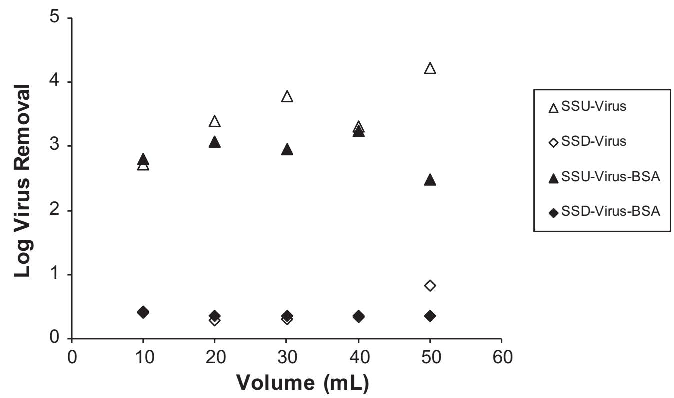

Fig. 7. Log virus removal for Omega 10 using the Amicon 8050 stirred cell. SSU and SSD refer to skin surface facing upstream (toward) and downstream (away) from the feed stream. Virus and virus-BSA indicate virus containing feed streams without and with spiked BSA.

ultrafiltration membranes. These parvovirus particles are around 20 nm in size, and are very similar in size to the MVM virus tested here. They noted that only 30 and 50 kDa membranes completely rejected the virus particles. In all of these studies the membranes were used in tangential flow mode with the filtration (skin) surface facing the feed stream.

The rejection specifications of ultrafiltration membranes are much broader than virus filtration membranes and also depend on the specific operating conditions used—which frequently are not explicitly noted. As indicated in Table 1, manufacturers publish specific log removal factors for their virus filtration membranes and the operating conditions. In the case of ultrafiltration membranes, however, rejection factors typically are given in terms of the lowest molecular weight dextran for which 90% (1 log) rejection occurs. Consequently, it is not surprising that previous investigators have observed detectable concentrations of 100 nm virus particles in the permeate from 300 kDa membranes and 20 nm virus particles in the permeate from 100 kDa membranes.

In this work however, less than 1 log removal of virus was observed using the Omega 10 membrane even though it is recommended by the manufacturer for concentration of proteins with molecular weights above 30 kDa. It is important to note that the Omega 10 membrane is designed to be operated with the filtration surface (skin surface) facing upstream in tangential flow mode and that the manufactures claims are based on operation under these conditions. Fig. 7 gives the results for additional experiments conducted using the Omega 10 membrane using a smaller 8050 stirred cell. As can be seen in the presence and absence of BSA, if the membrane is operated with the skin surface facing the feed stream, about 3 log removal of virus is observed. However, when operated with the skin side away from the feed stream the level of clearance is less than 1 log in agreement with the results for the larger 8200 stirred cell in Figs. 5 and 6.

Syedain et al. [11] investigated the capacity of Viresolve 180 (Millipore) virus filtration membrane which is designed for passage of proteins up to 180 kDa [21] and exhibits very high BSA passage. When membranes were run with the filtration surface away from the feed stream (as is the case here), the decrease in permeate flux was due to entrapment of small protein aggregates within the membrane support structure as is the case here. However, when the membrane was operated with the filtration surface facing the feed stream, these investigators indicate that there was also significant concentration polarization that can lead to the observed decrease in permeate flux.

Our BSA rejection studies indicate an apparent BSA rejection coefficient for the Omega 10 membrane is about 25% (see Table 2).

While the Viresolve 180 is designed to pass proteins with molecular weights up to 180 kDa, the Omega 10 has NMWCO of only 10 kDa. Thus it should be far more retentive to BSA. The effect of concentration polarization and consequent osmotic pressure difference created between the high protein concentration next to the membrane surface compared to the bulk feed solution will be amplified by the fact that very high protein concentration solution will be trapped within the membrane support structure (internal concentration polarization) as has been shown by Boyd and Zydney [28] in an earlier detailed study. This could also explain the low apparent rejection coefficient observed for the Omega 10.

The very low rejection of virus particles by the Omega 10 is probably due to the fact that, unlike the DV20, the support structure is very open allowing virus particles to collect in the fluid that is trapped in the pores of the support structure (see Fig. 1(d) and (j)). The presence of even a few relatively large pores in the skin layer, will then result in passage of these virus particle into the permeate. Fig. 7 provides further evidence to support this conclusion. Using the Amicon 8050 stirred cell, about 3 log removal of virus was obtained when the skin surface faced the feed stream.

Care is needed when using a stirred cell to determine LRV for asymmetric membranes. In general a high LRV indicates a virus retentive membrane. However a low LRV could be due to poor virus retention by the membrane or leakage of feed solution around the O ring. We are confident that leakage around the O ring is not the source of the low LRV observed for the Omega 10 membrane run with the skin surface away from the feed stream for the following reasons. The difference in LRV between the filtration surface facing the feed (about 3 log) and away from the feed (significantly less than one log) is about 3 log (see Fig. 7), while the permeate fluxes are the same. It is unlikely sufficient leakage of fluid around the O ring to cause a 3 log difference in LRV would not lead to a measurable change in permeate flux. BSA rejection data are in agreement with previous studies [28] that indicate when using an ultrafiltration membrane with the skin surface away from the feed, the observed rejection is significantly lower.

In an earlier study Boyd and Zydney [28] investigated rejection by Omega 30 and 50 membranes of polydisperse dextrans suspended in PBS. Direct flow filtration experiments were conducted in a stirred cell with the filtration surface towards and away from the feed stream. Experiments were also conducted using a sandwich arrangement where an Omega 30 and 50 membrane were placed together with the filtration surface forming the external surfaces of the composite membrane. For single membranes, dextran rejection coefficients were in agreement with the manufacturers specifications when the membranes were run with the filtration surface facing the feed stream. However when the membranes were run with the filtration surface away from the feed stream, rejection coefficients up to an order of magnitude less than specified by the manufacturer were obtained, in agreement with the BSA rejection results obtained here for the Omega 10 membranes. Boyd et al. point out that while the presence of a more open layer beneath a tight filtration layer has little effect on the observed rejection coefficient, the reverse is not true due to internal concentration polarization.

These results support the empirical practice that optimized asymmetric virus filtration membrane structures should have a larger pore size surface facing the feed stream that acts as an in-line prefilter removing any small aggregates present in the feed. Further, any internal concentration polarization effects will reduce any tendency for protein rejection. In the case of essentially symmetric virus filtration membranes, while the upstream surface and membrane structure will trap any aggregates present, the advantage is that there is no downstream filtration surface to protect. However as indicated by Fig. 3, a disadvantage of a symmetric structure is a much lower permeability. Interconnections between the membrane pores are essential to ensure the permeate containing product can flow around ‘blocked’ pores. This arrangement will maximize membrane capacity.

Table 3
Values of the parameter β for the four constant pressure blocking models.

Model	Assumptions	β
Standard blocking	Foulants deposit evenly along the pore walls causing a decrease in pore diameter, number of pores per unit area remains constant	1.5
Complete blocking	Solute completely blocks a pore	2
Intermediate blocking	Solute partially blocks a pore with a certain probability	1
Cake filtration	Solute forms a cake on the filtration surface of the membrane, which leads to an increase in resistance to filtrate flow	0

For asymmetric and symmetric membranes, the pores of the surface facing the feed stream should have a relatively narrow pore-size distribution and should minimize entrapment of virus particles in the membrane itself. For asymmetric membranes containing a thin skin surface that faces away from the feed stream, if the upstream surface has a larger pore size than the virus particles, entrapment of virus particles in the support structure next to the thin skin layer occurs. Optimization of the upstream prefiltration layer involves minimizing virus polarization yet maximizing flux. Our results (Fig. 7) indicate that the skin layer has to be defect free; otherwise, there is a risk of virus passage into the permeate.

The results of this work suggest the higher throughputs of composite asymmetric membranes and the lack of need to protect a thin skin layer of symmetric membrane may be combined by designing membranes where the retentive ‘layer’ is in the middle of the membrane and not at either surface. The benefits of such a structure could include protection of the upstream and downstream surfaces of the retentive layer and use of a thinner upstream prefiltration layer. This could lead to the fabrication of thinner retentive layers thus increasing throughput. In addition, since the downstream support structure will have little effect on membrane performance and does not perform any in-line prefiltration function, it can be relatively open again reducing the resistance to permeate flow.

Constant pressure direct flow filtration is often described in terms of four blocking models: standard blocking, complete blocking, intermediate blocking and cake filtration [10,29,30]. Hermia [29] has shown that all four blocking models have the form

$$\frac{d^2t}{dV^2} = \alpha\left(\frac{dt}{dV}\right)^{\beta} \quad (2)$$

where V is the permeate volume and t is time. Consequently, dt/dV is the reciprocal of the permeate flow rate and d^2t/dV^2 is the resistance coefficient defined as the rate of change of the instantaneous resistance to filtration with respect to permeate volume. The parameters α and β are constants. Values of β are given in Table 3 and may be used to identify the actual mechanism of flux decline during filtration. It is important to remember however, that these blocking models strictly apply to membranes containing cylindrical non-connected pores that are perpendicular to the membrane surface. Thus they may be used to obtain only qualitative information on the fouling mechanism for the membranes investigated here.

Eq. (2) indicates that a plot of $\log(d^2t/dV^2)$ versus $\log(dt/dV)$ should yield a straight line with a slope equal to β. Fig. 8 presents results for the membranes tested here for feed streams containing MVM and 1% BSA. Results for the RVRM, DV20 and Omega 300 membranes yield straight lines with a slope approximately equal to 2 indicating complete blocking. The straight lines through the data in Fig. 7 indicate a slope of 2. This suggests that aggregated protein species are completely blocking pores in the membrane structure.

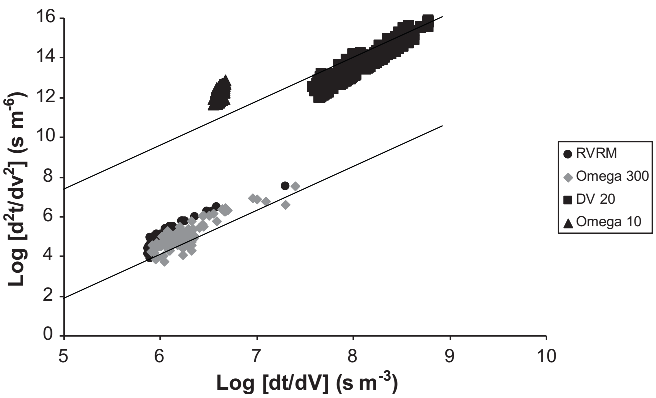

Fig. 8. Variation of resistance coefficient with the reciprocal of filtrate flow rate. The straight lines indicate a slope of 2.

Fig. 4 indicates that there is little flux decline in the absence of BSA, thus entrapment of BSA aggregates is most significant. The resistance coefficient is much higher for the DV20 in agreement with its much lower permeability. The resistance coefficients of the RVRM and Omega 300 membranes are similar in agreement with their similar water fluxes.

The results for the Omega 10 membrane indicate an approximately constant change in resistance coefficient. Further, the permeate flux appears to be approximately constant. In fact, Fig. 4 indicates that unlike the other membranes, the permeate flux decreases rapidly to a low value and then remains approximately constant. All four classical pore blocking models predict that the total resistance should depend only on cumulative permeate volume [11], which does not appear to be the case for the Omega 10 membrane. This observation and the determination that the Omega 10 membrane is highly retentive for BSA, suggest that internal concentration polarization and the associated increase in osmotic pressure between the high concentration protein solution trapped in the membrane support structure and the lower concentration in the bulk feed is the major cause of flux decline.

Taken together the results of the direct flow filtration experiments conducted here using virus filtration and ultrafiltration membranes with feed steams containing MVM and relatively low and high protein concentrations provide a number of insights into designing optimized virus filtration membranes. Filter capacity is maximized by minimizing flux decline due to adsorption of small aggregates of the product species (perhaps even dimers and trimers) on the membrane filtration surface. In this work we did not characterize the size of the aggregates present in our BSA containing feed streams. However Fig. 8 suggests flux decline due to complete blocking of the membrane pores is significant. In the case of asymmetric membranes, a slightly more open surface facing the feed stream can act as an in-line prefilter protecting the actual filtration surface from fouling by aggregating solutes. Pore interconnectivity in the support structure ensures that fluid can pass around 'blocked' pores as has been shown in an earlier study [31]. These asymmetric membranes display relatively higher throughput.

In the case of essentially symmetric membranes, the upstream surface and membrane structure may also trap aggregates present though there is no downstream skin surface to protect. Again pore interconnectivity is essential in order to ensure fluid can pass around blocked pores thus maximizing capacity. Symmetric structures display lower throughput compared to asymmetric structures though concerns regarding defects in the downstream skin surface compromising virus removal are eliminated.

Filter throughput is increased by maximizing membrane permeability (porosity). Membrane capacity is maximized by ensuring good pore interconnectivity. Membrane selectivity, i.e. maximizing virus rejection and protein passage will depend on maximizing virus rejection by the membrane surface in contact with the feed stream. The membrane pores should be uniform in size and similar or slightly smaller than the virus particles. This will maximize complete rejection of virus particles by the membrane's retentive surface, reduce entrapment within the membrane structure, and minimize the risk of passage through any larger pores (or defects) in the membrane structure.

4. Conclusions

Virus filtration experiments have been conducted using commercially available retrovirus and parvovirus filtration membranes, as well as ultrafiltration membranes with NMWCOs of 300 and 10 kDa. All membranes were operated in direct flow mode. For asymmetric membranes, the filtration surface formed the downstream membrane surface. Feed streams containing MVM and MVM plus 1% BSA were used to challenge the membranes. The results obtained here suggest that optimized asymmetric virus filtration membranes should contain a slight gradation of pore size, the larger pore size surface being in contact with the feed stream. This more open surface will act as a prefilter removing any aggregates that could foul the skin filtration (retentive) surface. For symmetric membranes, the upstream surface and the membrane structure may remove aggregates present, though there is no specific retentive surface to protect. For both asymmetric and symmetric membranes, extensive pore interconnectivity is essential to ensure the permeate flux is maximized by allowing fluid to flow around blocked pores. Our results suggest that alternative structures where the retentive layer is located within the membrane, may combine the advantages of symmetric and asymmetric structures. It is essential the membranes contain a very narrow pore-size distribution to ensure maximum rejection of virus particles and passage of product species. This is particularly important when designing virus filters to reject small parvovirus particles.

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